

CCA GAME DEVELOPMENT

KINEMATICS & FRICTION

Building a Custom Physics Engine

Module 2 – Session 6

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Today's Mission

By the end of this session:

01

Understand velocity and acceleration

02

Build movement using a custom physics system

03

Add friction to slow objects down

04

Compare direct movement with physics-based movement



GOAL

Make movement feel **smooth and controlled**, not robotic.

Why Does Movement Feel Robotic?

```
position += direction * speed * dt
```

Press RIGHT → → → → → →

Release key **STOP**

The movement changes immediately.

What if we want:



We need more than just **position + direction**.

A Better Movement Model

Instead of:

INPUT



POSITION

We build:

INPUT



ACCELERATION



VELOCITY



POSITION

Each concept has a different job:

● **Acceleration**

Changes velocity.

● **Velocity**

Changes position.

● **Position**

Tells us where the object is.

Velocity

Velocity describes:

How fast an object is moving and in which direction.

With Vector2:

```
velocity = pygame.Vector2(0, 0)  
velocity = pygame.Vector2(5, 0)
```

`velocity = (5, 0)`



5 pixels per second →

Velocity can change over time.

Acceleration

Acceleration changes velocity.

ACCELERATION

↓ changes

VELOCITY

↓ changes

POSITION

```
velocity += acceleration * dt
```

If acceleration points right:



the velocity becomes more and more rightward.

The Physics Update

Every frame, our physics engine does:

1 Calculate acceleration

2 Update velocity

3 **Apply friction**

4 Update position

5 Draw

REPEAT

THE CORE FORMULAS

```
velocity += acceleration * dt
```

```
position += velocity * dt
```



Runs every single frame

Direct Speed vs Acceleration

Direct Movement

```
position += direction * speed * dt
```

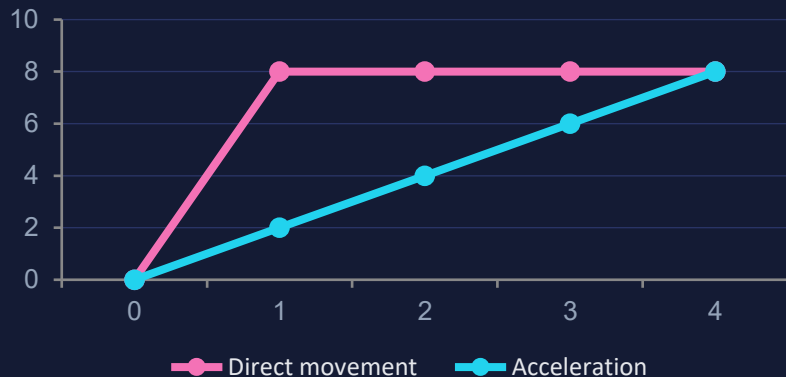
Press RIGHT: **speed** → instantly

Acceleration

```
velocity += acceleration * dt
```

Press RIGHT: 0 → 2 → 4 → 6 → 8 → ...

Speed over time (frames)



The Difference

Direct movement: speed changes immediately.

Acceleration: speed changes gradually.

What Happens When We Release the Key?

WITHOUT FRICTION

ACCELERATE



The object keeps moving.

We need a force that reduces velocity.

WITH FRICTION

FAST



SLOW



STOP

This is friction. Friction makes velocity gradually decrease.

Friction in Code

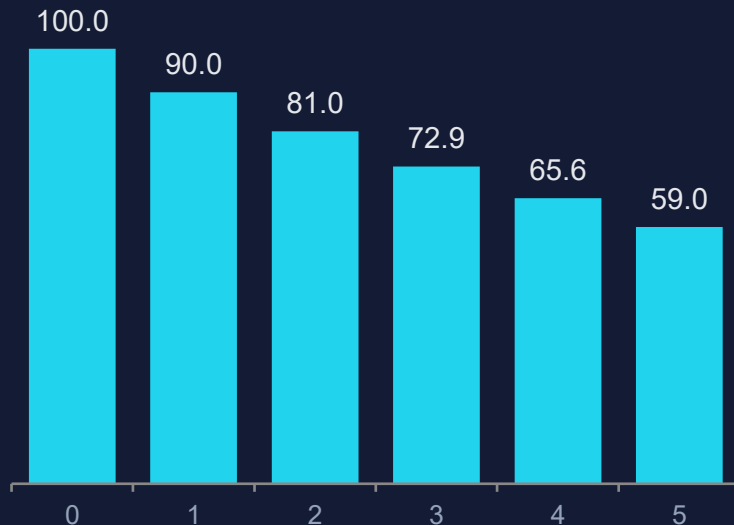
A simple physics model can reduce velocity over time:

```
velocity *= friction  
friction = 0.9
```

Smaller friction value → stronger slowdown

Closer to 1 → more sliding

Velocity after each update (friction = 0.9)



Custom Physics Engine

We now separate the systems:

1 INPUT

```
acceleration.x += 1
```

2 PHYSICS

```
velocity += acceleration * dt  
velocity *= friction
```

3 MOVEMENT

```
position += velocity * dt
```

4 DRAW

```
rect.center = position
```

Let's Build It



STARTING POINT

Start from the Session 5 Space Shooter.

github.com/annisafzhn/CCA.git



Find the Player movement code.



player.py

```
# Add
self.velocity = pygame.Vector2(0, 0)

# Create acceleration
acceleration = pygame.Vector2(0, 0)

# Update velocity
self.velocity += acceleration * dt

# Apply friction
self.velocity *= 0.9

# Finally
self.position += self.velocity * dt
```

Test & Observe

Run the game.

TEST 1

Hold **RIGHT**.

What happens to the velocity?

TEST 2

Release **RIGHT**.

Does the player stop immediately?

TEST 3

Change `friction = 0.9` to
`friction = 0.5`

TEST 4

Try:
`friction = 0.99`

Observe: How does friction change the movement feel?

Physics Challenge



Can you tune the physics?

CHALLENGE 1

Acceleration

Change the acceleration value.

Make the player:

Slow to accelerate

or

Fast to accelerate

CHALLENGE 2

Friction

Find a friction value that feels:

Smooth

but does not take too long to stop.

CHALLENGE 3

Compare

Create two versions:

Version A

High acceleration + strong friction

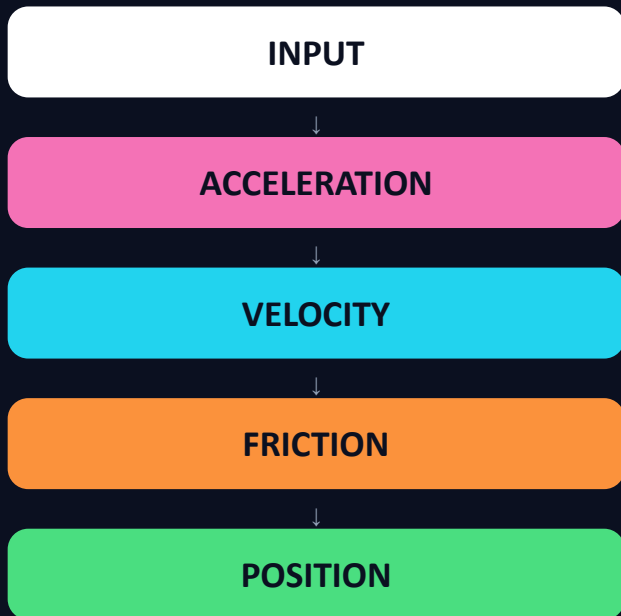
Version B

Low acceleration + weak friction

Which one feels different? Explain why.

Today's Big Ideas

We changed the movement model:



CORE FORMULAS

```
velocity += acceleration * dt
```

```
velocity *= friction
```

```
position += velocity * dt
```

Physics gives us control over how movement feels.

NEXT SESSION

Gravity & Jumping Mechanics

What happens when acceleration is always pointing down?

GRAVITY

→

JUMP

→

FALL

→

LANDING

See you in Session 7!

